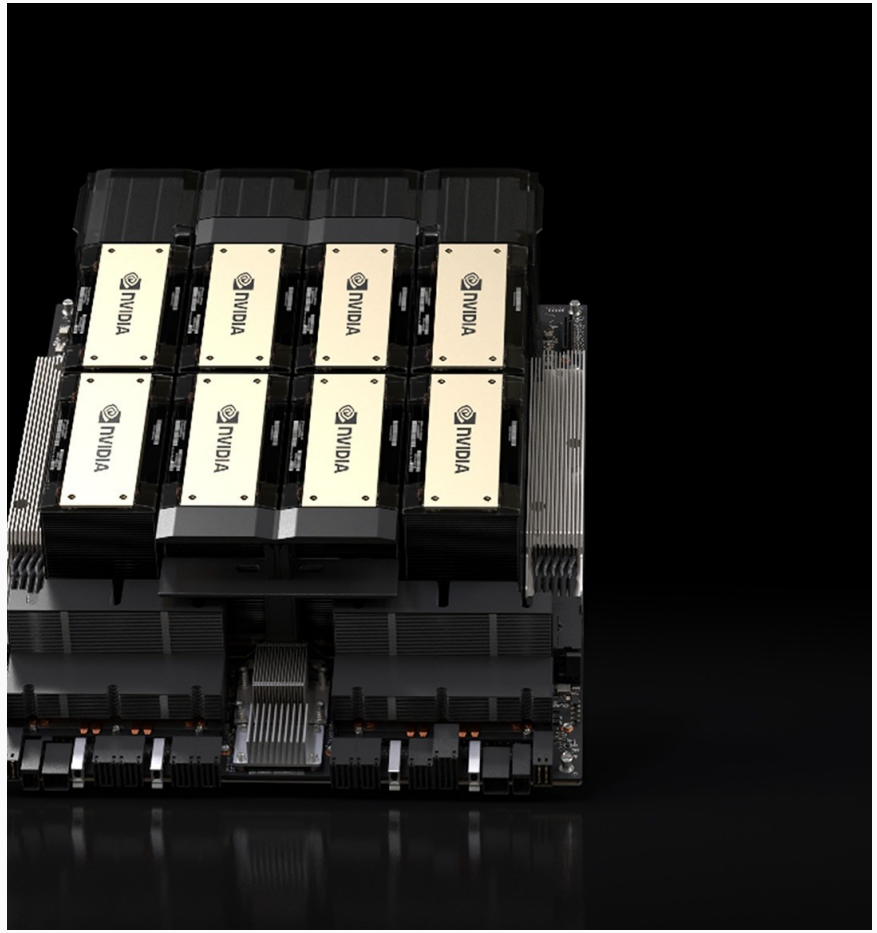
Qwen3-235B PD inference on 16 H200s

VNUHCM - TripleU

12 October 2026



NVIDIA HGX H200 · product reference image

Local evidence · job 22415 · organiser accuracy and official re-run pending

Measured result

1,912.30

good output tokens / second

97.44%

within both SLAs



Job 22415 · N48 · 60s warmup / 120s measurement / 30s grace

The latency gate

4,000-token prompt

200 generated tokens



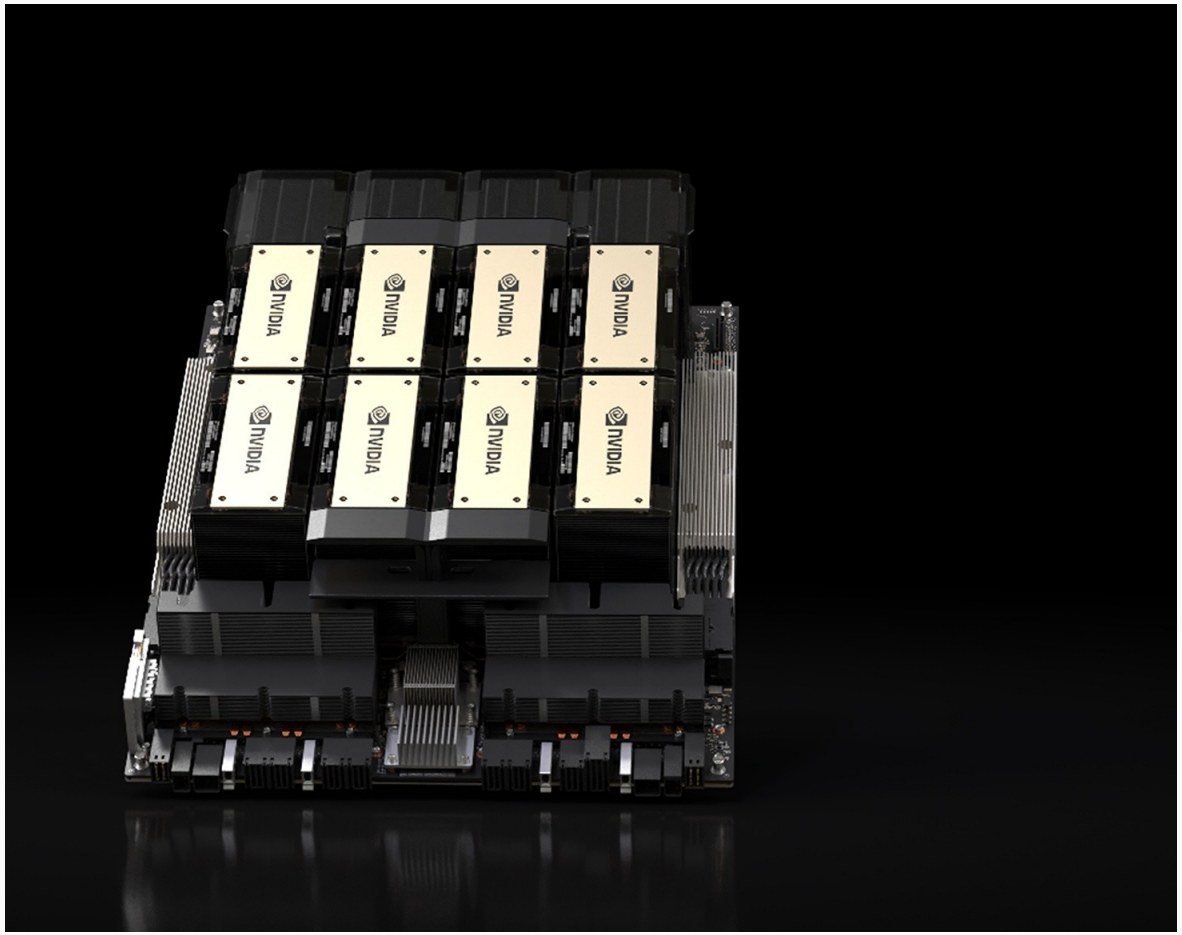
Time to first token ≤ 3,000 ms

Mean ITL ≤ 20 ms

≥ 90% of requests must pass both. Errors stay in the denominator.

Official workload: 4000 input / 200 output tokens; TTFT ≤ 3000ms and mean ITL ≤ 20ms

Two nodes. Sixteen H200 GPUs.



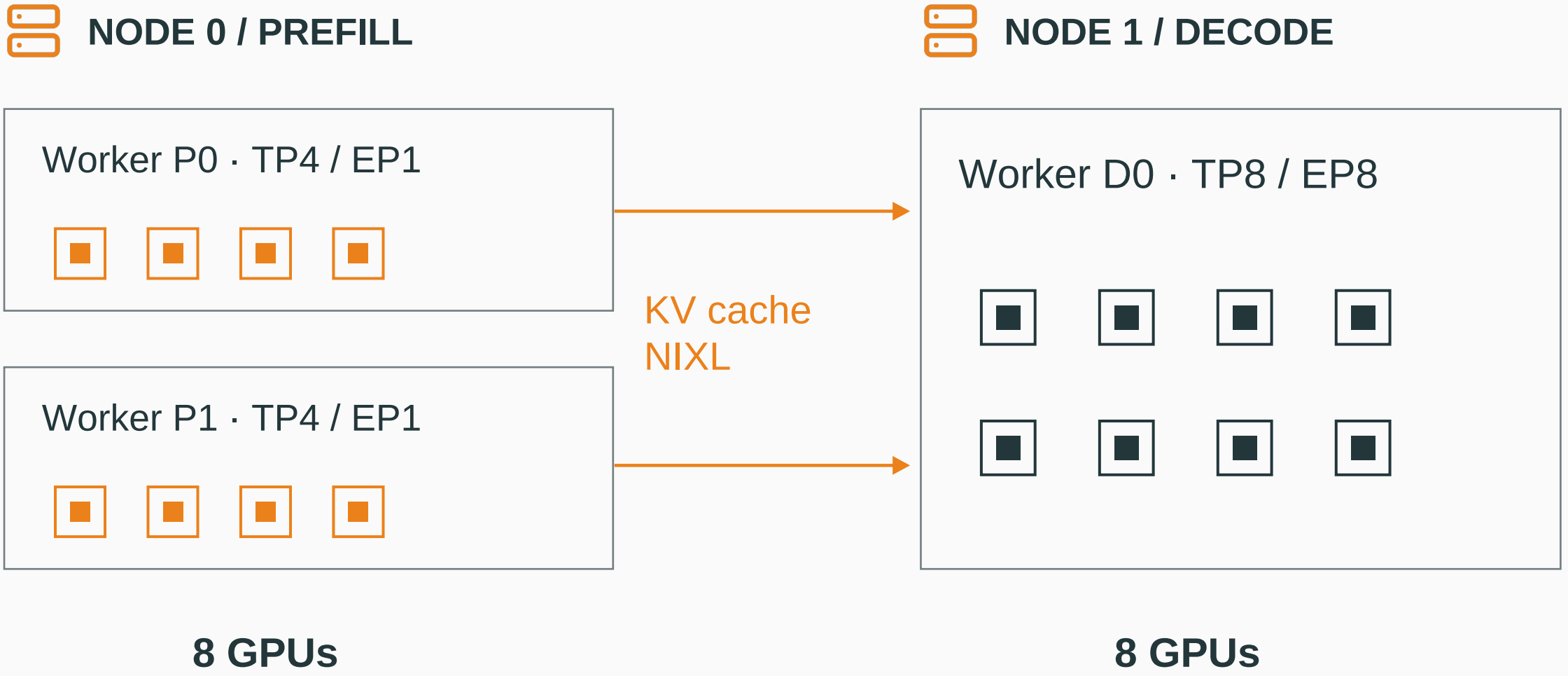
16 GPUs

InfiniBand NDR

MODEL	Published Qwen3 FP8
KV CACHE	BF16, verified in worker logs
SERVER	SGLang 0.5.17 / NIXL 1.3.2
CLIENT	AIPerf 0.12.0

Contest allocation: 2 × HGX H200; photo: NVIDIA H200 product page, not the contest installation

Winning topology: TP4 prefill → TP8 / EP8 decode



P8-TP4_D8-TP8-EP8 · two P/D paths · every scored request uses the gateway

From the provided layouts to the selected split

07

PREFILL



4 workers × TP2

DECODE

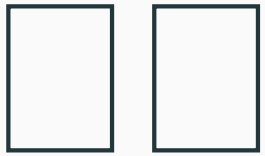


2 workers × TP4/EP4

1,297.02

09

PREFILL



2 workers × TP4

DECODE

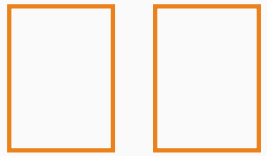


2 workers × TP4/EP4

1,666.01

Selected

PREFILL



2 workers × TP4

DECODE



1 worker × TP8/EP8

1,912.30

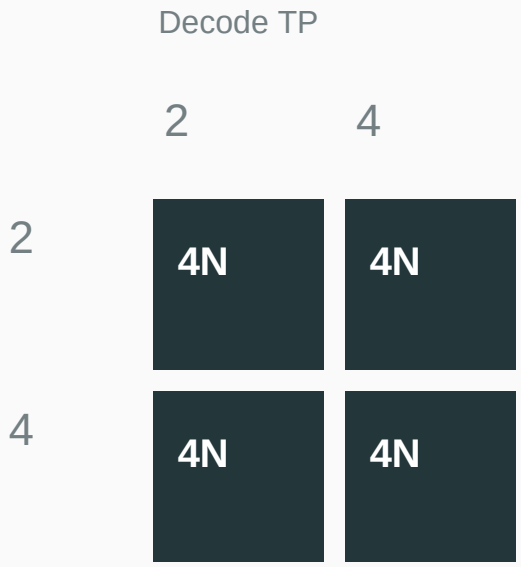
Long local runs: 22197 / 22219 / 22415; N32 / N40 / N48 respectively

Seventeen usable layouts measured

P8 + D8 GPUs



P4 + D12 GPUs



P12 + D4 GPUs



All layouts: N16 / N32 / N40 / N48

14 exact originals + 3 explicit EP2 replacements; incompatible TP8/EP1 originals remain labelled

One job: deployment followed by the official benchmark

Deploy → initialize both P/D paths → AIPerf at N48

Official client timeline

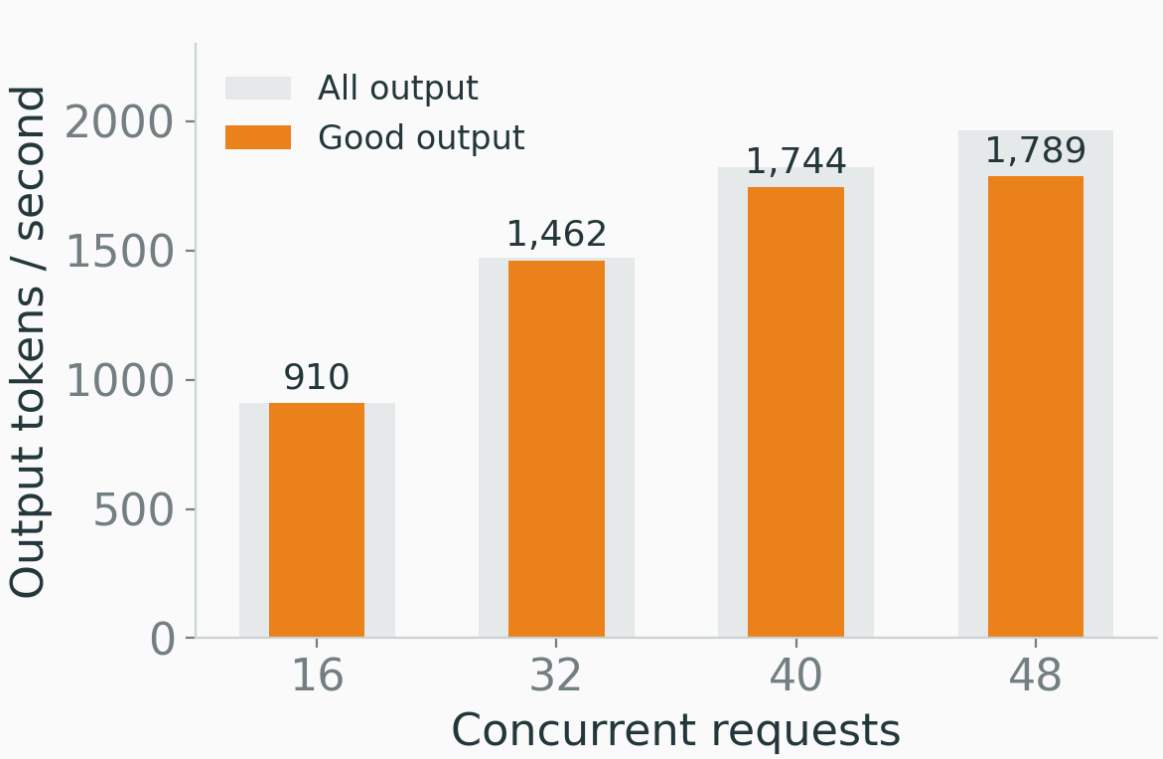


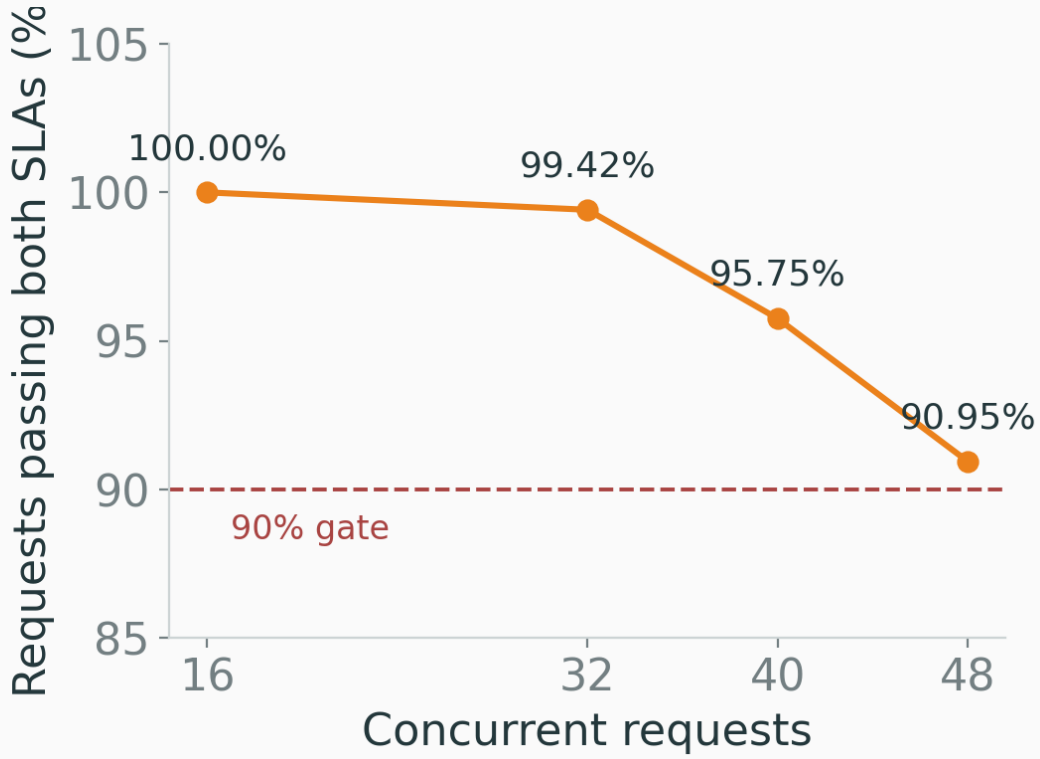
Only this window contributes to goodput.

Closed loop: completion immediately triggers the next request, sustaining N48.

Primary submit.sbatch has one N48 invocation; the official command and seed are unchanged

Load selection: N48 passes the short sweep

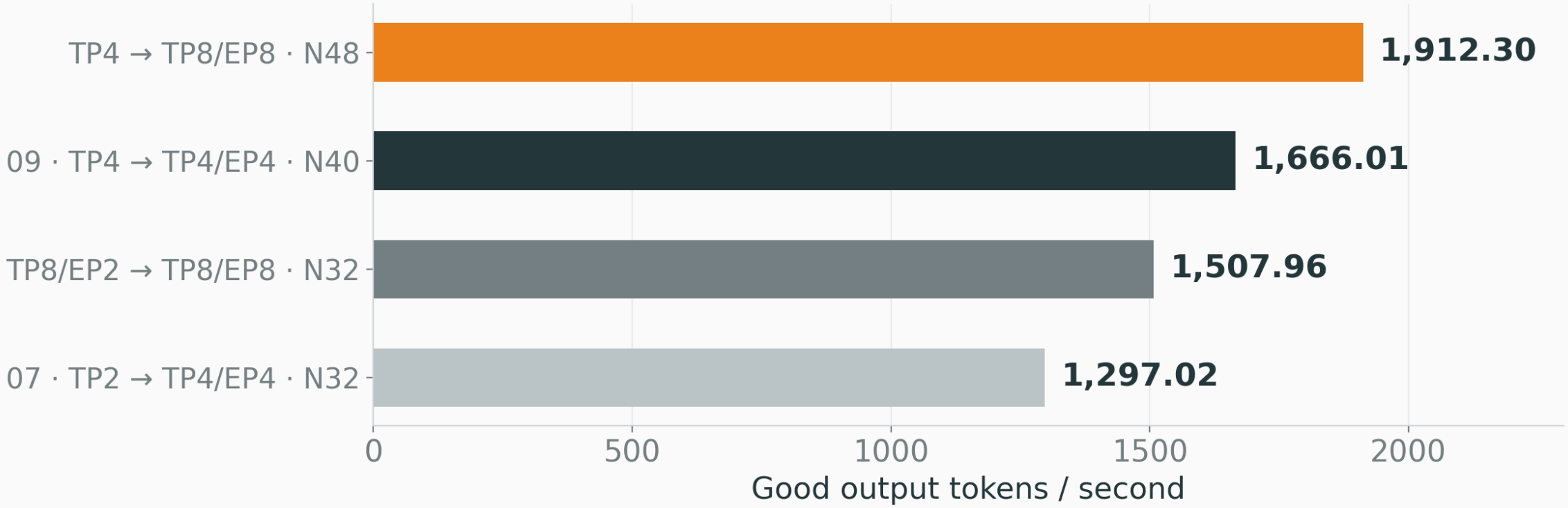




Short N48: 1,788.65 good tokens/s · 90.95% SLA

Job 22415 · exploratory 10 / 20 / 30 runs; confirmation uses a separate 120s window

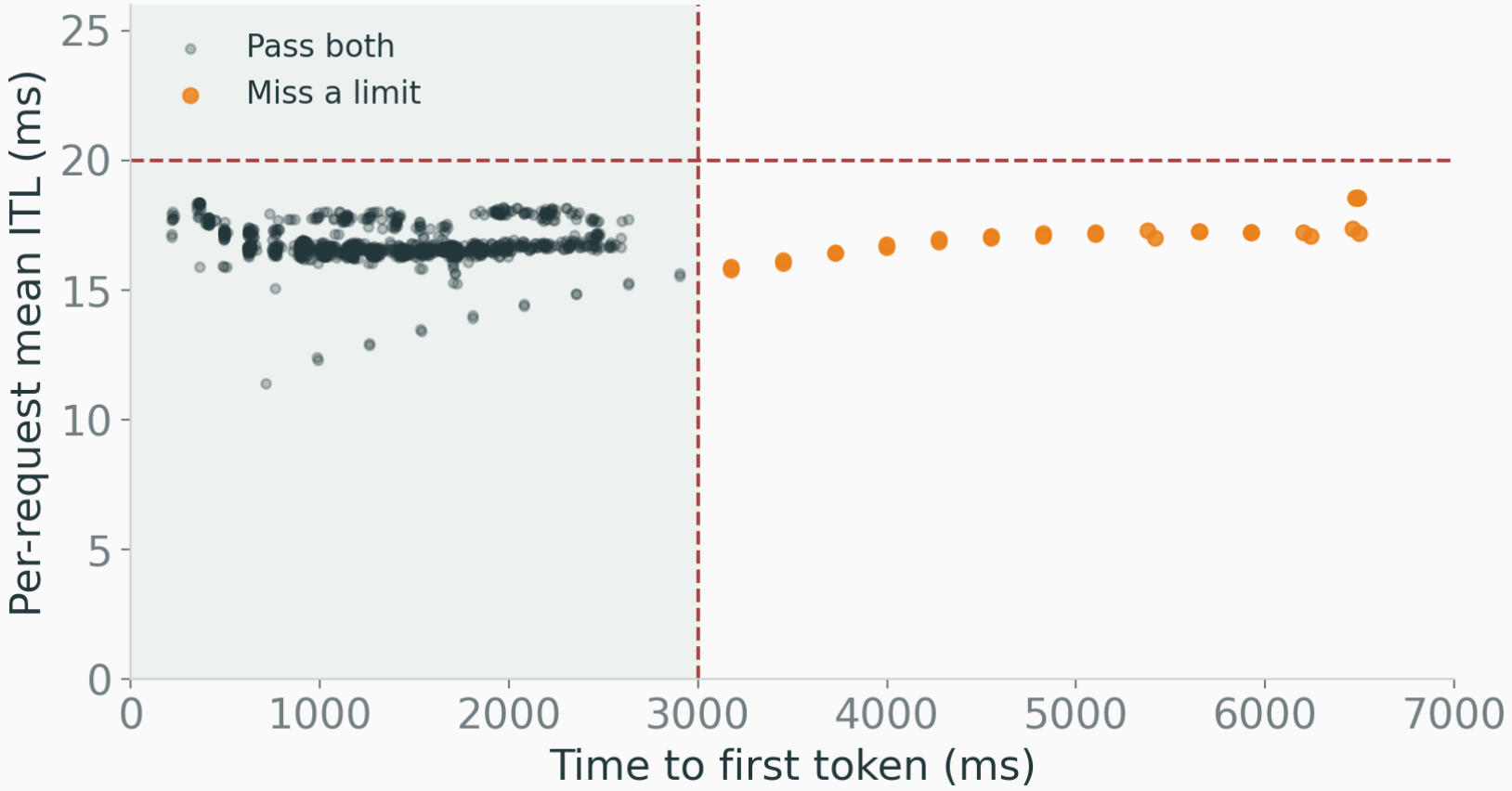
Long-run comparison: +14.78% over the previous best



Topology, concurrency and transport changed; this is an end-to-end gain.

All bars: passing local 60 / 120 / 30 confirmations on the same 16-H200 platform

Per-request evidence: 1,180 of 1,211 pass both limits



97.44%

31 miss a limit
0 errors
0 cancellations

TTFT mean 1,482 ms
ITL mean 16.79 ms

Original job-22415 profiling JSONL; counts independently recomputed from both latency limits

Cold setup failed before the benchmark began

Before / job 22778

480s

Pair warmup timed out.
No benchmark export.

After / job 22809

COMPLETED

Workers ready at 269s.
All P/D paths warm at 303s.
Four sweeps + long N32.

Failed jobs remain in the experiment history; they do not count as measurements.

Unpatched 22778 failed; equal-TP-patched 22809 completed. This does not isolate every timeout cause.

FP8 block alignment requires a distinct EP2 variant

TP8 / EP1

1536 ÷ 8 = 192



A partial block remains.

TP8 / EP2

1536 ÷ 4 = 384

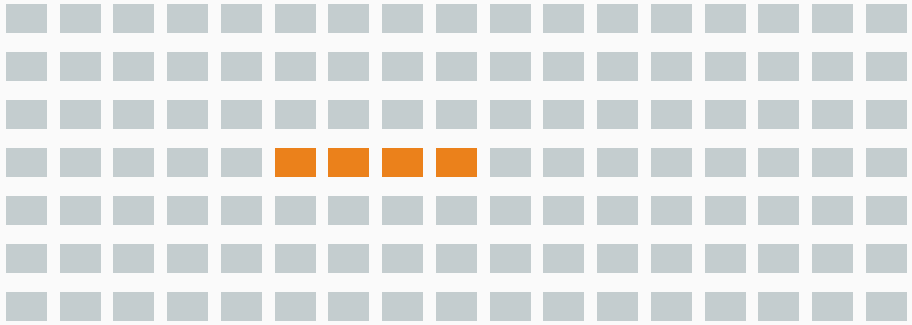


Three complete 128-wide blocks.

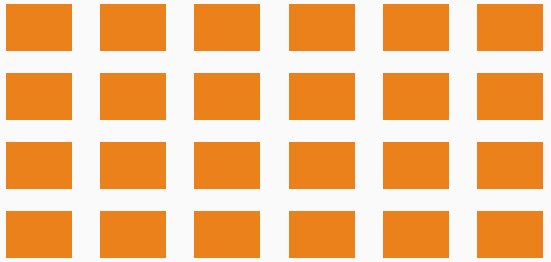
MoE intermediate dimension 1536; FP8 block_n = 128; all 128 experts and published weights retained

NIXL: describe the request, not the whole KV pool

Whole-pool plan



Request-scoped plan



403.8 million

source entries / rank

188

coalesced entries / request

CPU byte-range cases: 720 heterogeneous TP + 936 equal TP8; both patches served GPU benchmarks.

Geometry example: TP8 source, 94 layers, 4,000 contiguous tokens; not every TP4→TP8 transfer has 188 entries

The primary job deploys the winning stack

Reconstruct under a dedicated account

```
bash setup.sh --check
sbatch submit.sbatch
```

1	**Recorded images**	Published FP8 checkpoint; offline tokenizer.
2	**Winning layout**	Two TP4 prefill workers; one TP8/EP8 decode worker.
3	**Enabled patch**	Source hash verified by every worker before launch.

AI_Task/job_scripts/submit.sbatch · package-relative paths · read-only NIXL module overlay

Latency and accuracy are separate acceptance checks

Latency

97.44%

Passing local measurement
Required: at least 90%

Accuracy

Pending

Organiser full-set checks
Required: ≥99% of reference
FP8 weights remain unchanged

Full-set accuracy threshold: 99% of the organisers own FP8 reference score; no accuracy pass is claimed

Declared Basic configuration

TP4 prefill → TP8 / EP8 decode

1,912.30

good output tokens / second

N48 · 16 H200 · 97.44% SLA · request-scoped NIXL descriptors

Local performance evidence; accuracy and official organiser re-verification pending

Appendix · the final repaired TP8 / EP2 layout

SHORT SWEEP	GOOD TOKENS/S	SLA	GATE
N16	933.50	98.15%	Pass
N32	1,393.62	93.02%	Pass
N40	1,624.14	89.50%	Fail
N48	1,708.25	86.06%	Fail

Long N32: 1,507.96 good tokens/s · 95.38% SLA

Job 22809 · equal-tp8-request-descriptors-v1; short sweeps and long confirmation are labelled separately

Appendix · evidence and reconstruction

PRIMARY ENTRY	`AI_Task/job_scripts/submit.sbatch`
DECLARED POINT	P8-TP4_D8-TP8-EP8 / N48
SOURCE EVIDENCE	`AI_Task/logs/measured/22415/`
TEXT PATCH	`AI_Task/job_scripts/nixl.patch`
REBUILD METHOD	`AI_Task/job_scripts/README.md`

Team work: topology, load selection and NIXL integration.

Visual theme: Metropolis (CC BY-SA 4.0) · reference image: NVIDIA · icons: Lucide (ISC/MIT)